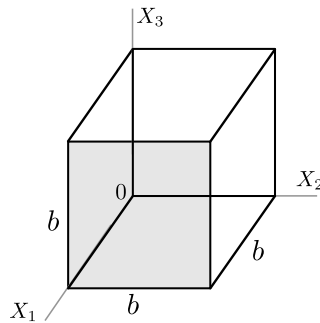


02MECH - Makeup test

20.01.2023

Question 1

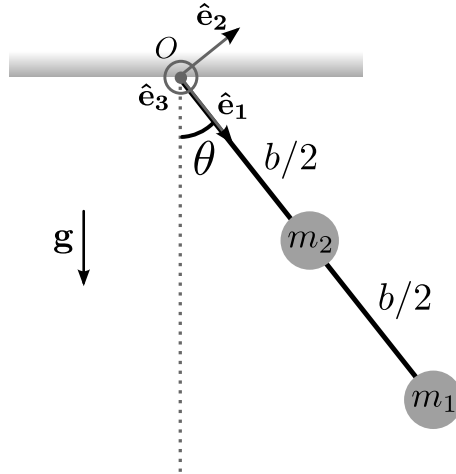


A homogeneous cube of density ρ , mass M , and equal side lengths b is given as shown in the figure.

- (a) Calculate the inertia tensor $[\mathbf{J}]$ by using $J_{ij} = \int \rho(r) (\delta_{ij} \sum_k x_k^2 - x_i x_j) dV$ by letting the corner O be the origin and three adjacent edges lie along the coordinate axes X_1, X_2, X_3 .
- (b) Write down the components of center of mass vector $\mathbf{a}$ with respect to X_i axes.
- (c) By using parallel axes theorem $I_{ij} = J_{ij} - M(\delta_{ij} a^2 - a_i a_j)$ calculate the inertia tensor $[\mathbf{I}]$ when we choose the origin as the center of mass.
- (d) What are the principle moments of inertia I_i ? Without calculating, can you predict what principle axes associated with I_i look like? Try to draw them on the figure.

Each option is worth 0.3 points - Total: 1.2 points

Question 2



m_1 and m_2 are fixed on a rigid rod of length b . $\hat{e}_i$ are the body coordinates. The mass of the rod is negligible.

- Calculate the inertia tensor $[I]$ of the rigid body by using $I_{ij} = \sum_{\alpha} m_{\alpha} (\delta_{ij} \sum_k x_{\alpha,k}^2 - x_{\alpha,i} x_{\alpha,j})$.
- Calculate the angular momentum vector L .
- Write down Newton's second law for rotations.
- Find the square of the angular frequency of small oscillations ω^2 if the pendulum swings in a plane.

Each option is worth 0.3 points - Total: 1.2 points

Question 3

For a given central force $f(r)$ the orbit of a particle of mass μ can be calculated by integrating the following differential equation,

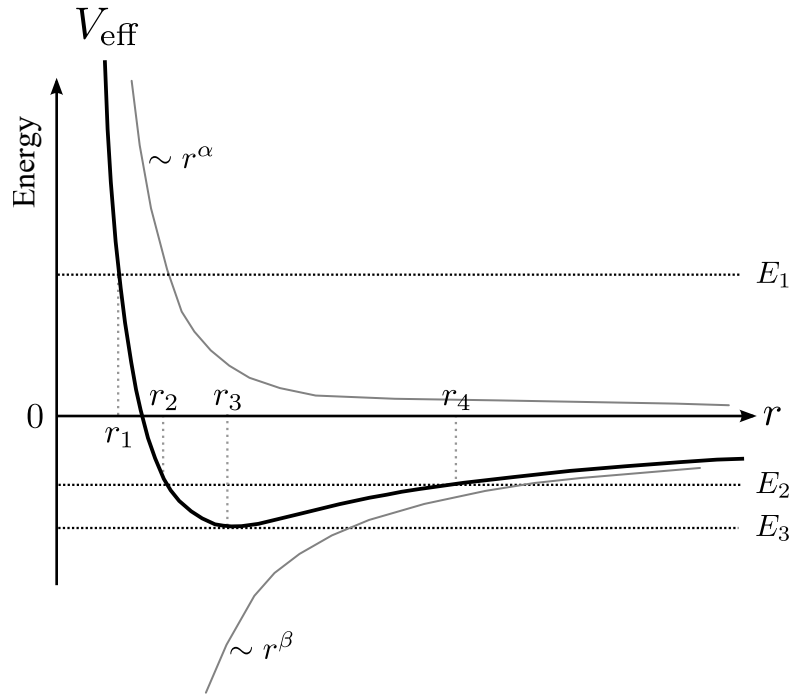
$$\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{\mu}{l^2} r^2 f(r)$$

where l is the angular momentum of the particle.

- (a) Find the force law for a central-force field that allows a particle to move in an orbit given by $1 + \varepsilon \cos \theta = \alpha/r$, where $\alpha \equiv \frac{l^2}{\mu k}$ and $\varepsilon \equiv \sqrt{1 + \frac{2El^2}{\mu k^2}}$ are constants. Write down the central force in terms of k and r .
- (b) The effective potential energy is given as $V(r) = U(r) + \frac{l^2}{2\mu r^2}$, where $U(r)$ is the potential energy. First show that $U(r) = -\frac{k}{r}$, and then prove that the particle follows a circular orbit when $E = V_{\min}$, where $V_{\min}$ is the minimum effective potential energy. (Hint: First think of what ε has to be for a circular orbit. Then calculate the r value making $V(r)$ minimum. Show that ε satisfies circular orbit condition when you set $E = V_{\min}$ for ε .)

Each option is worth 0.6 points - Total: 1.2 points

Question 4



In the figure the effective potential energy of a particle moving in an inverse-square-law central-force field $f(r) = -k/r^2$ is given. One of the gray curves behaving proportional to r^α is the centrifugal potential energy. The other curve behaving proportional to r^β is the potential energy in the central-force problem.

- Given that the speed is expressed by $v^2 = \dot{r}^2 + r^2\dot{\theta}^2$ in polar coordinates, and that the angular momentum is $l = \mu r^2\dot{\theta}$, write down the total energy of the particle in terms of $\dot{r}$, l , μ , r and k .
- Which term in total energy is called the centrifugal potential energy, and how the effective potential V_{eff} is defined? According to this, determine α and β in the figure.
- Describe briefly how a particle would possibly move in this central-force field when it had total energies E_1 , E_2 and E_3 .

Each option is worth 0.4 points - Total: 1.2 points